1. For acoustic simulation results:

- T1w, segmentation, ROI visualization in Axial, sagittal, and coronal slice cut through the slice of centroid of sgACC (target)

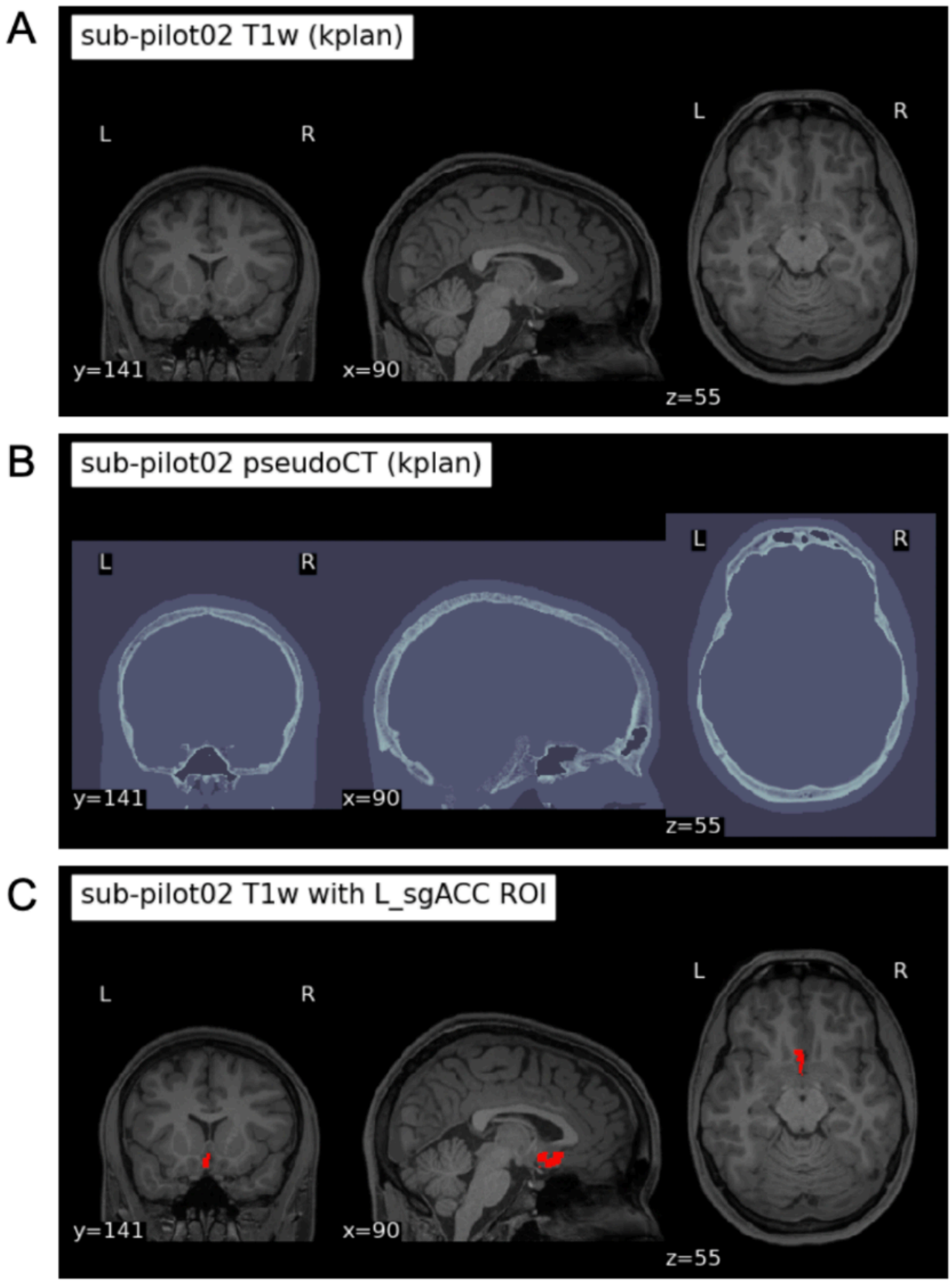
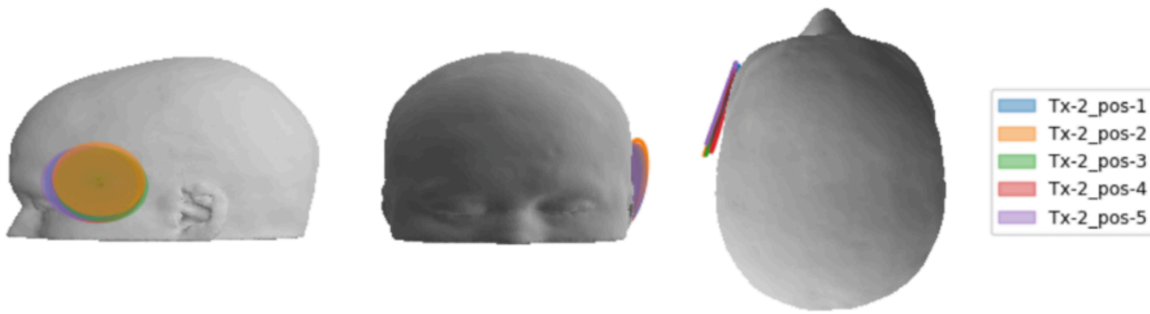


Figure 2. *Individual anatomy, skull modeling, and sgACC target definition.* ***A.*** *High-resolution T1-weighted anatomical image.* ***B.*** *PETRA-derived pseudo-CT used for skull modeling and acoustic simulation.* ***C.*** *Left sgACC ROI (BA25) transformed into individual space.*

Head mesh with all 5 transducer positions to run simulation:



Display as coronal/axial/sagittal overlays on the T1w image:

- Pressure map (slices including one set of cut through centroid of target, and a few other slices)

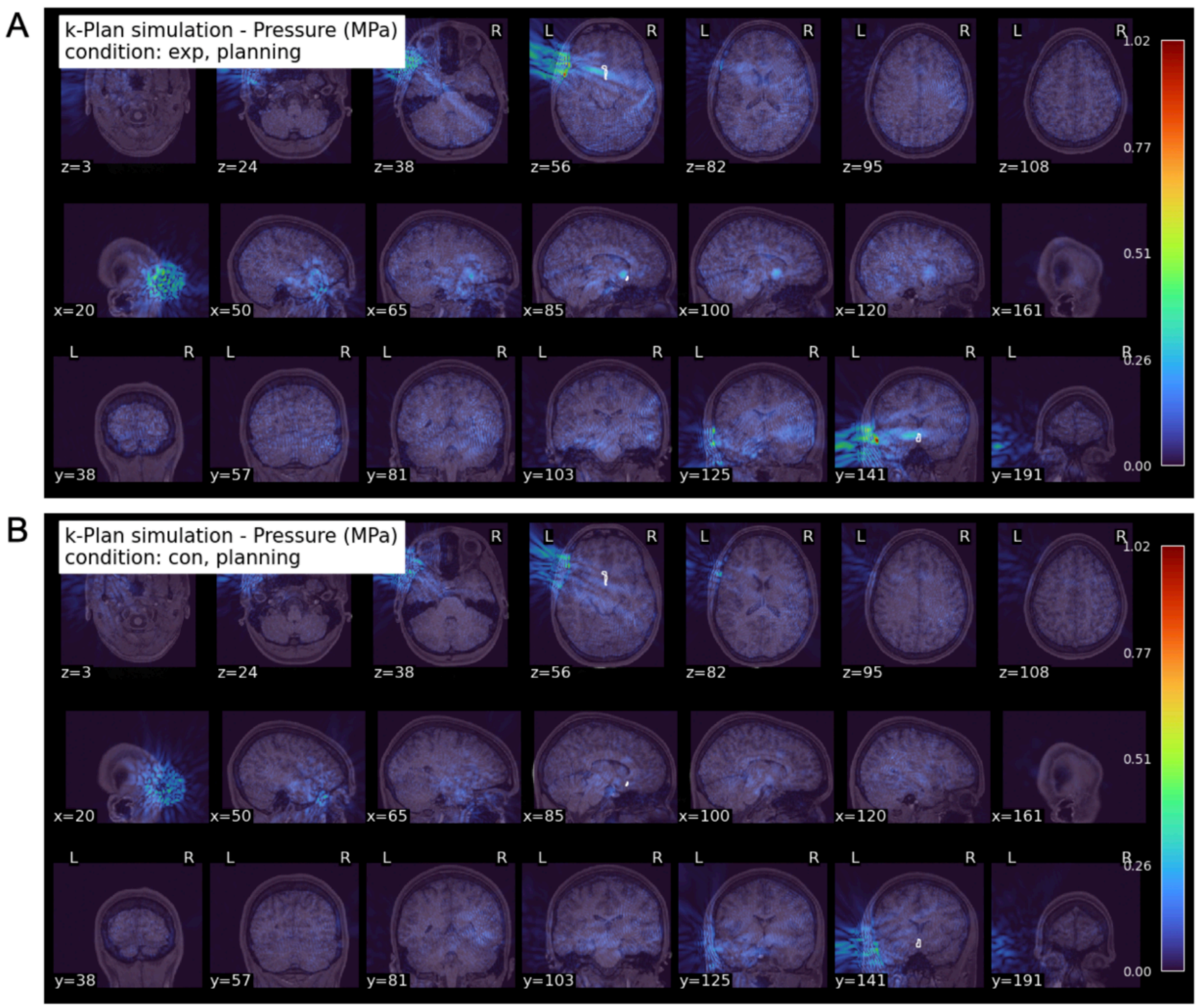


Figure 4. *Whole-head acoustic pressure distributions for protocol-1. Whole-head pressure maps derived from individualized k-Plan simulations for protocol-1 are shown in mosaic view for an example participant.* ***A.*** *Focused experimental condition.* ***B.*** *Defocused control condition. Both conditions use identical stimulation parameters and output levels, illustrating comparable global acoustic exposure while differing in spatial focusing. Color scales are identical across panels for appropriate comparison, and sgACC is marked as contour map in white line.*

- Temperature map:

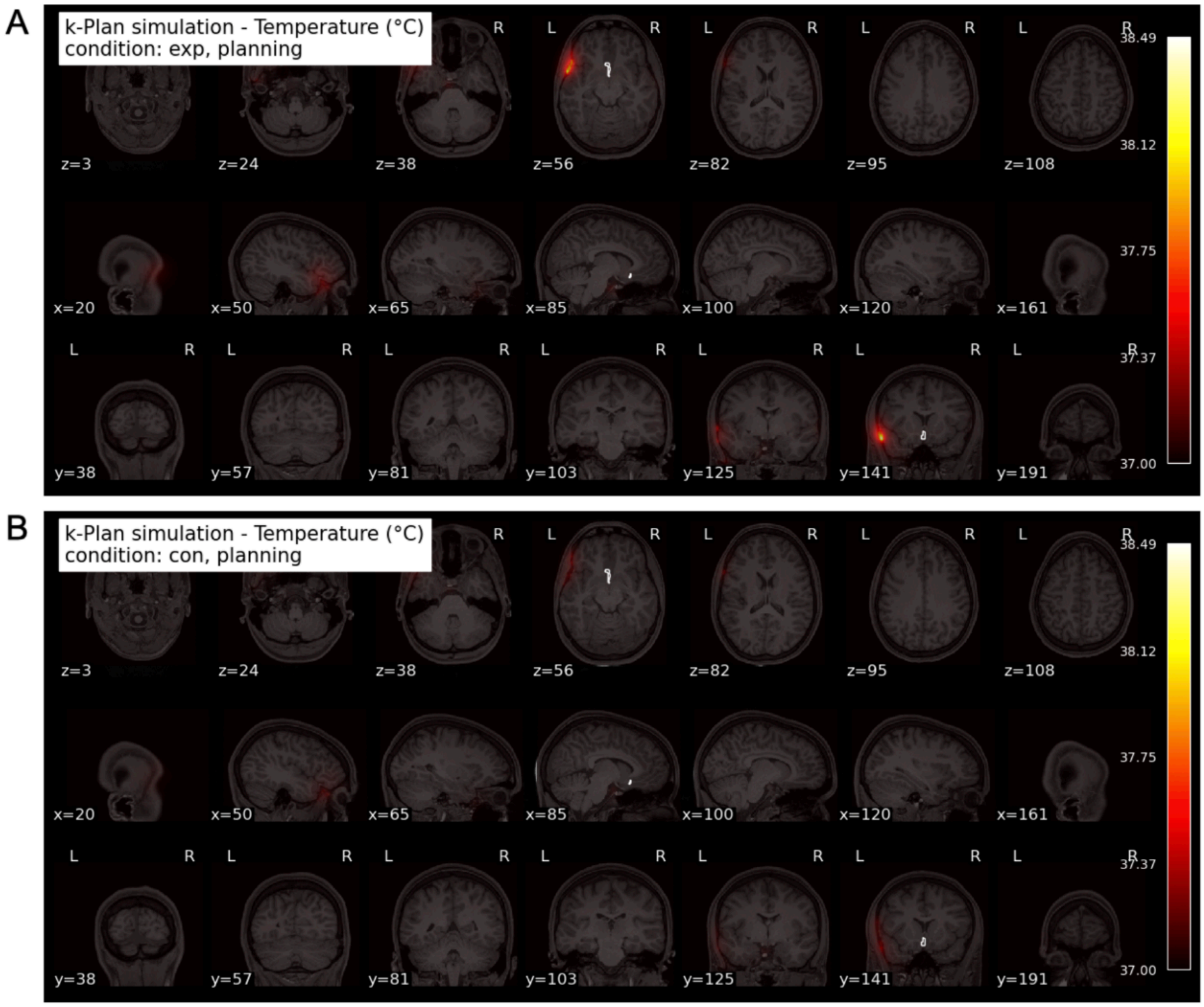


Figure 6. *Assessment of thermal safety using simulated temperature rise. Simulated temperature distributions for protocol-1 across whole head.* ***A.*** *Focused experimental condition.* ***B.*** *Defocused control condition. Temperature rise remained <2°C for both conditions across the head, and color scales are identical across panels for comparison.*

2. For post-hoc:

- Finalized DBSCAN clustering code to choose the best represented positions for the whole stimulation session in each hemisphere L&R.
- Head mesh with heatmap of all actual positions recorded during stimulation.
- Summary plot of maximum displacement from the reference position. Report mean ± SD of positional drift (in mm and degrees) of all Tx positions during TUS, like this:

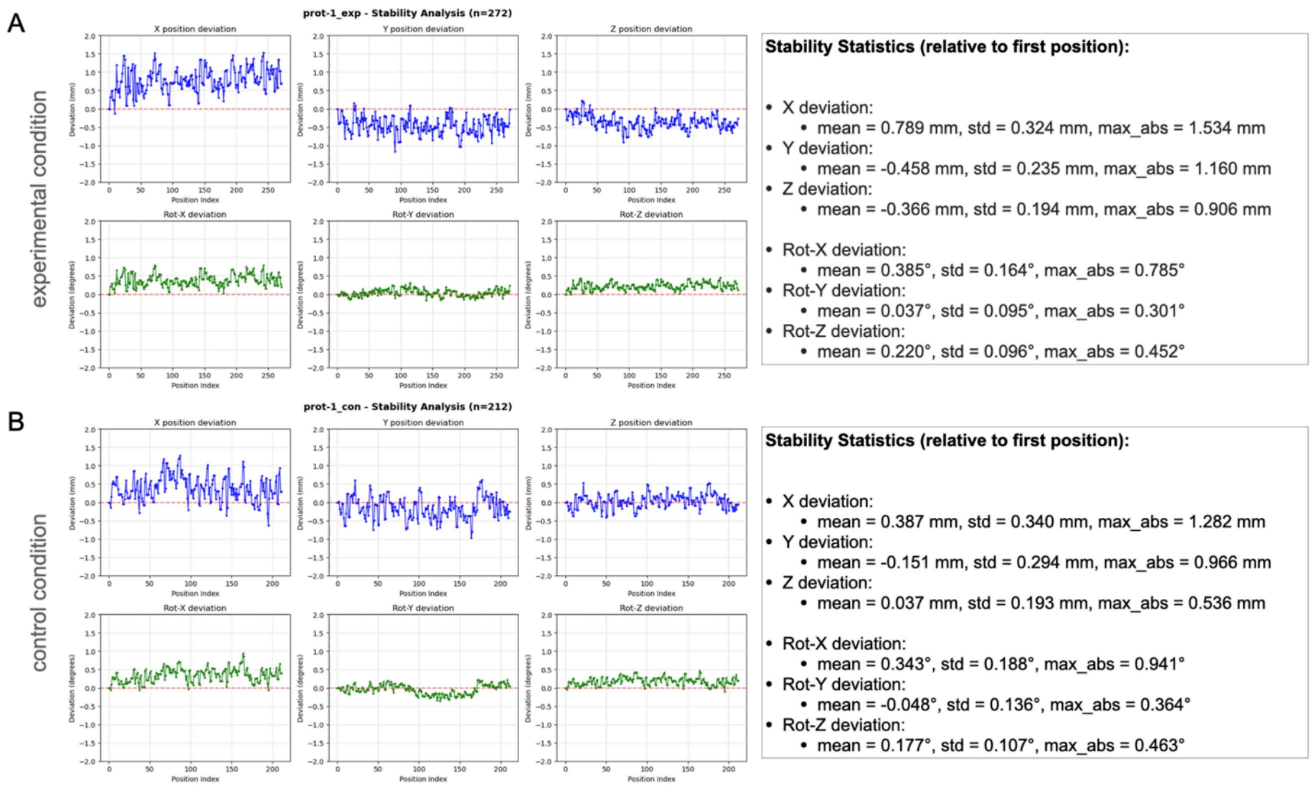


Figure 8. *Transducer position stability across representative runs.* ***A.*** *Translational and rotational deviations of the transducer position across repeated runs for the focused experimental condition, expressed relative to the first recorded position.* ***B.*** *Corresponding deviations for the defocused control condition. Deviations are shown separately for translational (x, y, z; mm) and rotational (pitch, yaw, roll; degrees) components. Summary statistics (mean, standard deviation, and maximum absolute deviation) are displayed alongside each plot to quantify run-to-run placement stability.*

- Show -3dB focal volume overlapping with sgACC mask: Compute the –3dB pressure half-maximum ellipsoid from the simulation and show its overlap with the BA25 mask. Report: focal spot dimensions (axial × lateral FWHM in mm), Dice overlap coefficient or % of focal volume within BA25 mask, distance from focal centre to BA25 centroid (mm). This is the key targeting precision figure.
- Show -3dB actual with -3db planned contour comparing actual vs planned focal volume.
- Histogram of Tx positions variabilities by group and explain how we choose the medoid Tx positions to do post-hoc.
- Plot actual vs planned. Tx positions, target pressure/ temperature focal zone (like below so easily compare)

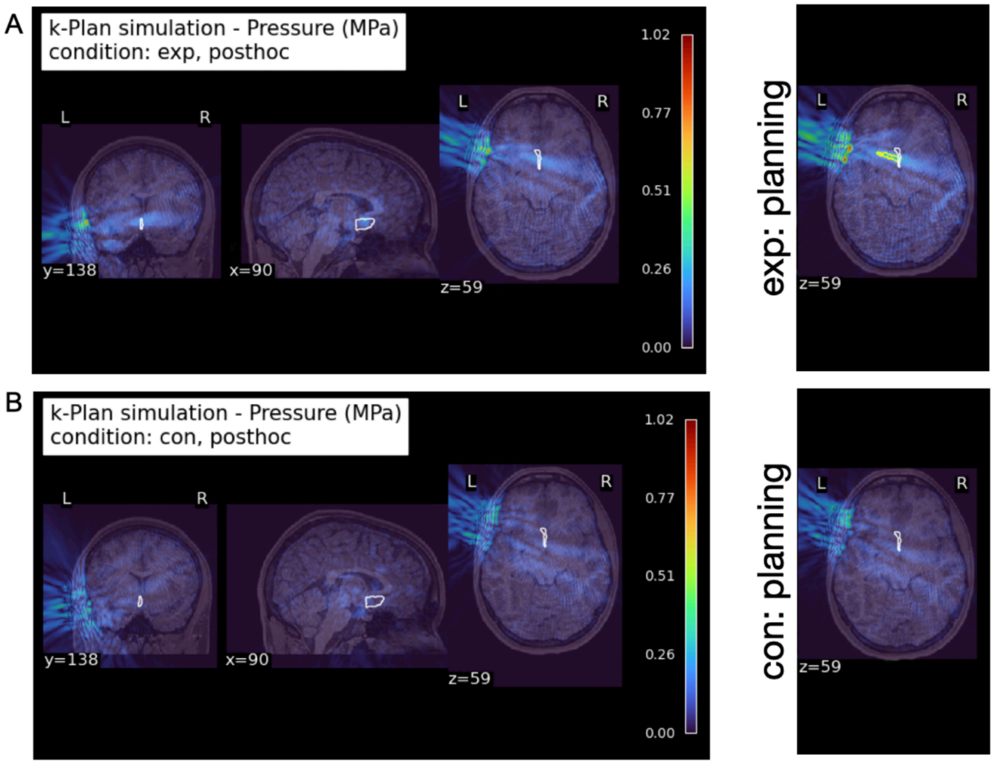


Figure 16. *Focused views of post hoc acoustic pressure at the sgACC target. Zoomed sagittal, coronal, and axial slices centered on the left sgACC illustrate the spatial relationship between the simulated pressure field and the target ROI for the experimental (**A**) and control (**B**) post hoc conditions. Pressure maps (MPa) are shown using identical scaling across conditions, including simulated pressure maps from prospective planning (right).*

- Do the same with Temperature, compare post-hoc vs planned temp: